NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Operating Systems (CS-3009) ■ Autumn-2022 ■ End Sem ■ 2 Pages
■ UG, 5^{th} Sem ■ FM: 50 ■ 3 hrs ■ Answer ALL

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1. (a) What is the basic role of a medium-term scheduler in process scheduling? [2]

   (b) Does one process thrashing affect only itself, or does it have an effect on the rest of the processes in the system as well? Explain briefly. [2]

   (c) Explain how multi-programming helps in achieving improved throughput and CPU utilization. [2]

   (d) What are the different states of a Process? Explain in brief with a neat diagram. [2]

   (e) Differentiate between Multitasking OS and Multiprocessing OS with their advantages and disadvantages. [2]

2. (a) Consider a pre-emptive scheduling policy called *Least Used First*(LUF). The scheduler selects the process/job that has used the least CPU time so far. For tie breaking the scheduler considers avoiding context switch then FIFO order of the ready queue. Once a job is selected as per LUF, it is allotted a time quanta of 2 units within which it can't be pre-empted, even by a newly arrived job. Assume one context switch time is 0.1 time unit. Consider 4 purely CPU-bound processes with their arrival time and burst time measured in *time units*, as mentioned below.

Job	Arrival Time	Burst Time
A	0	25
B	15	25
C	25	5
D	40	5

   Draw the Gantt chart showing the execution schedule of these processes and calculate the waiting time and response time of each job. The response time is the time between arrival and the time the job is first scheduled. At each scheduling point, mention the CPU time used so far by the available jobs and the content of the ready queue clearly. Justify if this algorithm can lead to starvation, considering multiple processes? [6]

   (b) Consider the reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. Simulate the operation of LRU and OPTIMAL page replacement algorithms assuming 5 frames, initially free. Calculate the Hit and Miss ratio. [4]

3. (a) Consider Dining Philosophers with N philosophers and N chopsticks. Assume L philosophers are left-handed and R philosophers are right-handed. Left-handed philosophers pick up their left chopstick first then the right chopstick. Right-handed philosophers pick up their right chopstick first then the left chopstick. $N = L + R$. Write the synchronization code, using semaphores, for the right-handed and left-handed philosophers. [4]
   Explain if deadlock can happen in each of the following cases: [3]

   - L=0 and R=N.
   - L=N-1 and R=1.

   (b) Contiguous file allocation introduces storage management problems in terms of external fragmentation. Discuss two alternatives to contiguous allocations in terms of their pros and cons. [3]

4. (a) Consider a system with 3 types of resources: X, Y, and Z; and 3 processes: P0, P1, and P2. Table 1 presents the current system state, and is self explanatory.
   There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state: [4]

   i. REQ0: P0 requests 0 units of X, 0 units of Y, and 2 units of Z.
   ii. REQ1: P1 requests 2 units of X, 0 units of Y, and 0 units of Z.

   Check if the above requests can be granted or not.

Table 1: State of Resource Allocation

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	0	6	2	0
P2	2	1	1	3	3	3

(b) What happens on a context switch? Context switches should happen frequently. Give your comments in for and against this statement. [3]

(c) Explain how operating system supports inter-process memory protection so that a process cannot access the address space of any other concurrently executing processes. [3]

5. (a) Consider a single-level paging scheme with a TLB. Assume no page fault occurs. It takes 20 ns to search the TLB. The TLB hit ratio is 50 % and the effective memory access time is 170 ns. Calculate the main memory access time. [3]

(b) Given five memory partitions of 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 212 KB, 417 KB, 112 KB, and 426 KB (in order)? Which algorithm makes the most efficient use of memory? [3]

(c) It is believed that increasing the RAM of a computer typically improves its performance. Is it true or false. Give your reasoning. Consider a virtual address space of 32 bits and page size of 4kB. System is having a RAM of 128kB. Then what will be the ratio of Page Table and Inverted Page Table size if each entry in both the tables is of 4 Bytes. [4]